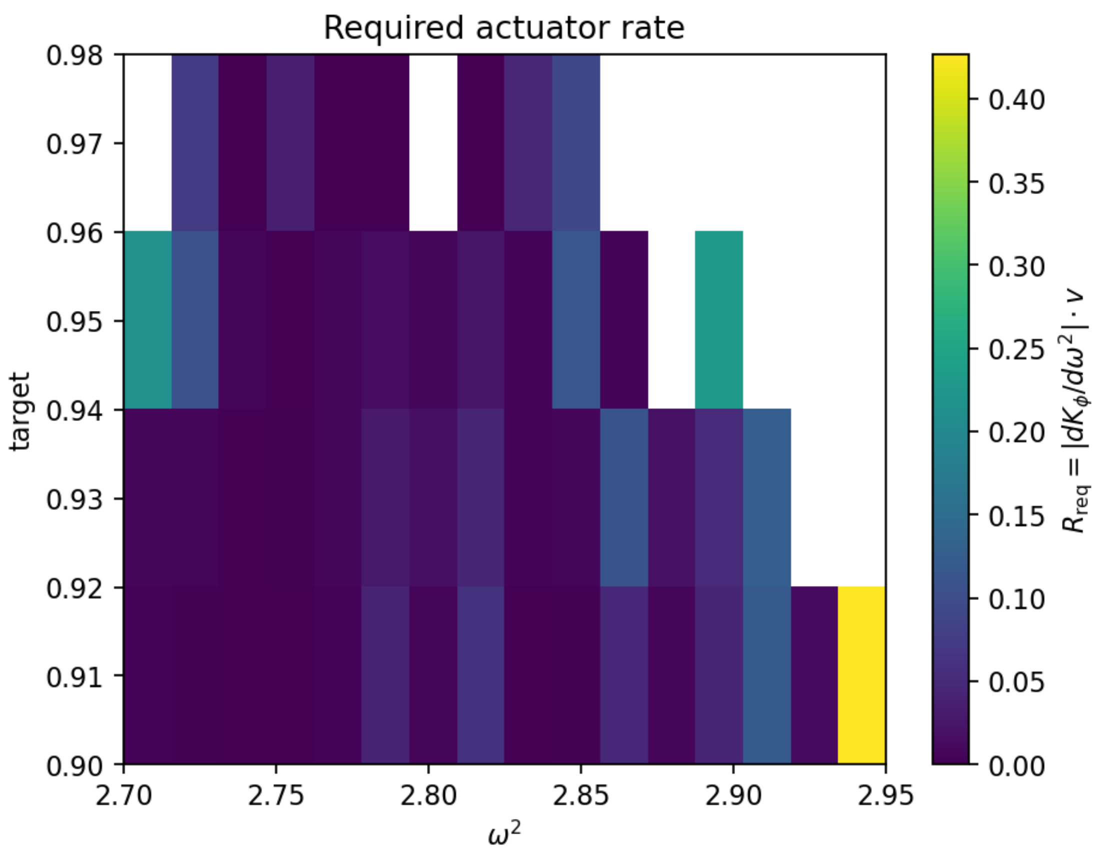
# Phase-24 — Lock-Loss Forecasting

- Drift input: `outputs/phase23/p23_drift_points.csv`
- Assumptions: $R = 6.0$ (max $dK_\phi/dt$), $v = 0.01$ ( $d\omega^2/dt$ )

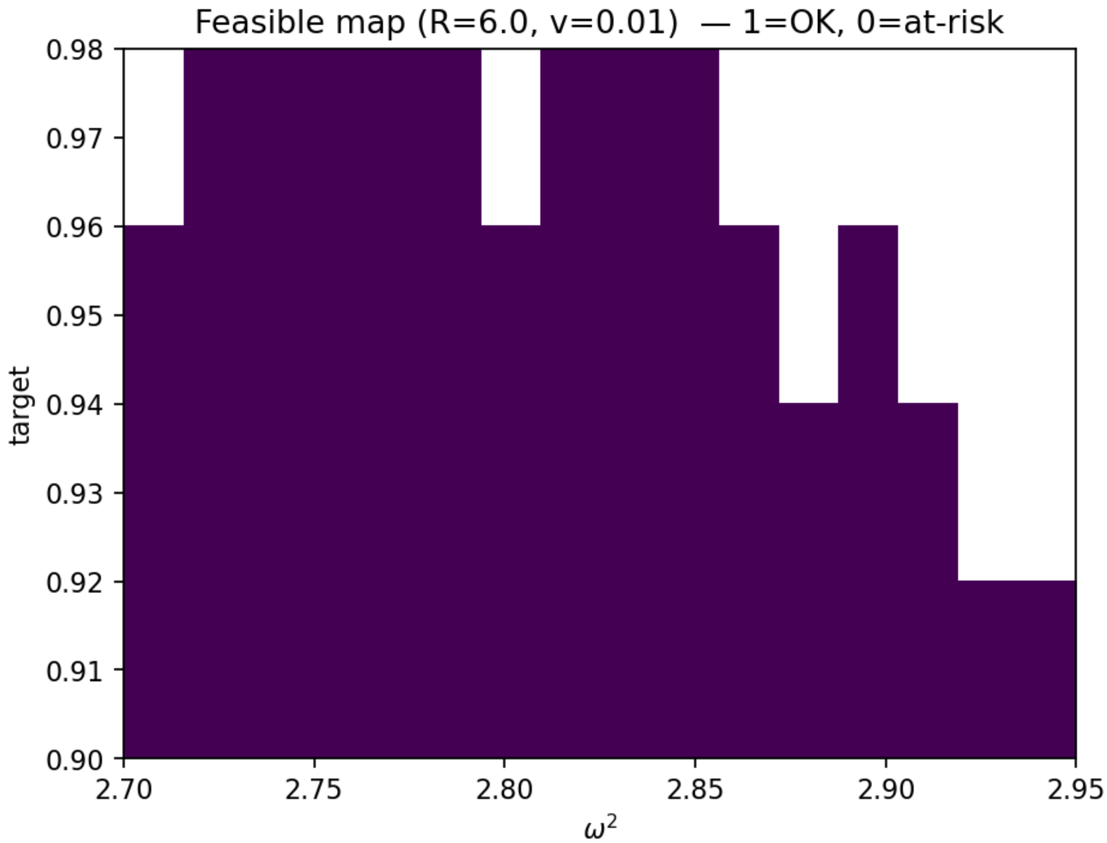
## Method

We estimate the required actuator rate per point as $R_{\text{req}} = |dK_\phi/d\omega^2| \cdot v$. If $R_{\text{req}} > R$, the controller cannot keep up and the lock is at risk.

## Required rate $R_{\text{req}}$ heatmap



### Feasible vs at-risk map (given R, v)




### Recommended Ops (from windowed sweep)

- Window: $\omega^2 \in [2.74, 2.92]$
- Targets: $\{0.94,\ 0.98\}$
- Selected actuator cap: $R = 6$
- Expected sweep speed: $v = 0.01$
- Criterion: 95% feasible over the window